# Jiaxin (Jeremiah) Pan

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## Research Interests

Physics undergraduate working on AI for Science, generative sampling, and interpretable crystal descriptors. Interested in reliable machine learning, language-model reasoning, and music-related AI; combines scientific computing with piano and choral training.

## Education

2023–2027 **B.Sc. in Physics (with Honours) — First Class Honours expected**
The Chinese University of Hong Kong, Shenzhen
GPA: 3.61/4.0. Four-year degree with a required research thesis. Admitted to Economics in 2023; completed Year 1 foundation courses before transferring to Physics in 2024.

2026 (Spring) **Visiting Student**
University of Auckland · New Zealand
Completed coursework in Physics, Computer Science, and AI, including COMPSCI 361 and COMPSCI 220.

2025 (Summer) **Summer Visiting Student**
University of California, Berkeley Extension · USA
Completed coursework in physics and neuroscience.

## Current Research

Ongoing **Data-Free Generative Sampling of the Two-Dimensional XY Model**
Undergraduate Thesis, CUHK-Shenzhen
Supervised by Prof. Kai Zhou. Provisional portfolio title.

- Contributed to adapting Stochastic Path Sampling (SPS) to periodic XY angles using energy and forward/backward path probabilities without equilibrium training data; evaluated paths with path-space independence Metropolis–Hastings (MH) correction.
- Completed reproducible Python/PyTorch training-budget experiments across three seeds and three generation-step settings on Slurm-managed GPUs; used Wolff sampling as an independent validation reference.
- Evaluated physical statistics, effective sample size per sampling second, and rejection streaks; found consistent early gains and seed-dependent later changes. Scope: 4×4, one temperature; no established thermodynamic critical behavior or advantage over Wolff.

Ongoing **Interpretable Crystal Representations for AI-Assisted Materials Design**
HSSRLM — AI for Science — Working paper
Hierarchical crystal descriptors with T5 (Text-to-Text Transfer Transformer) bidirectional text/descriptor infrastructure; broader data validation and model evaluation ongoing.

- Contributed to evidence-aware CIF (Crystallographic Information File)-to-descriptor preprocessing preserving provenance and known/inferred/missing information.
- Contributed to candidate validation, node/linker/topology connectivity checks, explicit failure reporting, and descriptor-level duplicate tracking.
- Contributed to limited structure-equivalence audits on three publicly sourced metal–organic frameworks, examining information lost by coarse descriptors.

Descriptors omit full atomistic structures; rule checks do not prove physical validity. Co-first author; manuscript in preparation for planned submission to the Annual Meeting of the Association for Computational Linguistics (ACL). Not yet submitted.

## Additional Research Experience

2026 (Spring) **Remote Research Assistant**
INSAIT · Bulgaria / Remote
Under Dr. Jinjin Gu: technical analysis of large language model reasoning and post-training, including Surgical Post-Training (SPOT), knowledge retention, and regularization; prepared research syntheses and contributed to seminars.

2025 (Fall) **Guided Research on Van der Waals Integration and 2D Electronics**
CUHK-Shenzhen, Prof. Qi Qian's Lab
Hands-on fabrication and characterization of $MoS_2$ devices using dry transfer and van der Waals integration; studied metal–semiconductor contacts and polarity modulation. Conducted literature review and prepared an end-of-term report.

2025 (Spring) **Computational Physics Course Project**
CUHK-Shenzhen
Group study of the quantum harmonic oscillator using finite-difference discretization and matrix eigenvalue methods; compared numerical and analytical results, examined convergence, and synthesized the final report in LaTeX. Technical note on Zenodo (2026).

## Manuscript and Conference

Submitted **Mathematics self confidence and STEM major choice in an undergraduate survey**
Reinvention: an International Journal of Undergraduate Research
Sole author; awaiting editorial assessment. Presented by invitation at the CUHK-Shenzhen Undergraduate Research Conference (URC2026), 31 January 2026. URC2026 paper and slides (Zenodo). Same URC2026 paper on SSRN.

## Teaching

Oct–Dec 2025 **Undergraduate Student Teaching Fellow (USTF)**
CUHK-Shenzhen School of Science and Engineering
Provided tutoring through the Student Help Room, a form of USTF service, for CSC1005 (Introduction to Computer Engineering), CSC1001 (Introduction to Computer Science), and PHY1001 (Mechanics), 23 October–12 December 2025.

## Musical Training and Performance

Piano Passed the Chinese Musicians Association amateur piano Grade 8 examination.

Choral singing Tenor in CUHK-SZ Chorus (Fall 2026), Auckland University Student Choir (Spring 2026), and School of Music Choir, CUHK-SZ (choir class, Fall 2025). Performed in two public Auckland concerts and *Winter Carols* at CUHK-SZ. Performance record; concert recordings.

## Honours and Certification

2023–2026 Full-Tuition Scholarship, CUHK-Shenzhen; awarded based on the 2023 Gaokao.

2024–2026 Dean's List: School of Management and Economics (AY2023–2024); School of Science and Engineering (AY2024–2025 and AY2025–2026). The 2026 certificate is pending issuance.

2024–2025 Shaw Overcoming Adversity Award, CUHK-Shenzhen Shaw College.

2025 HCIA-AI: Huawei Certified ICT Associate — Artificial Intelligence; valid through December 2028.

## Skills and Languages

Computing Python, PyTorch, NumPy, SciPy, Matplotlib, Git, LaTeX; GPU experiment workflows through Slurm.

Languages English: TOEFL iBT 100 (October 2025); Mandarin: native. Choral pronunciation training in German, Cantonese, Latin, Maori, and Japanese.